

CLOAKD

Dress your best.

Final Submission

Team Name: CLOAKD

Proposed Level of Achievement: Apollo 11

Table of Contents

- Motivation
- Vision
- User Stories
- Scope of Project
- Proof of Concept
- Features
    - User Account Authentication
    - Virtual Wardrobe
    - Outfit Builder
    - Occasion and Weather Filter
    - Outfit Calendar
    - Share Outfits and Community Feed
    - Settings Page
    - Profile Page
    - Statistics Page
    - Cost-Per-Wear Tracker
    - Streak System
    - Wardrobe Utilisation Score
    - Trending Outfits
- Testing
    - Testing Approach
    - Unit Tests Edge Cases
    - Manual Test Cases
    - Devices Tested
    - Known Limitations
- Timeline and Development Plan
- Database Schema
- Tech Stack
- Project Log
- Project Video
- Project Poster

Motivation

For most people, us included, getting dressed is stressful. You stand in front of a full wardrobe and somehow still feel like you have nothing to wear. CLOAKD reframes that daily moment as something expressive and enjoyable instead of a chore and removes the anxiety from picking out an outfit.

Most people wear fewer than a third of the clothes they own. The average piece of clothing gets worn just 7 times before it's thrown away and 40% of what we buy never gets worn at all. People keep buying new clothes and feeling like they constantly have nothing to wear not because they lack options, but because they can't see what they already have. CLOAKD solves a **visibility problem, not a shopping problem.**

Fast fashion is one of the most polluting industries in the world, but telling people to "buy less" doesn't work. CLOAKD takes a different approach by making it fun, social and rewarding to get more out of the clothes you already own. The cost-per-wear tracker and wardrobe utilisation score turn sustainability into a game rather than a guilt trip.

We also wanted to build something that brings people together around style in a way that feels real and also to learn how to take an idea from nothing and turn it into something real while picking up software engineering skills along the way by building something we actually care about.

Vision

CLOAKD will be a mobile app for iOS and Android, built with React Native and Expo. It lets users digitise their entire wardrobe and receive daily outfit suggestions built entirely from clothes they already own, filtered by real-time weather and occasion.

The app is designed around a simple daily loop: open the app, get a suggestion tailored to today's weather and occasion, log what you wore and watch your wardrobe come alive through stats, streaks and community sharing.

Ultimately, we envision CLOAKD as the app people reach for every morning, the one that changes how they relate to the clothes they already own, reduces mindless overconsumption and makes getting dressed something enjoyable.

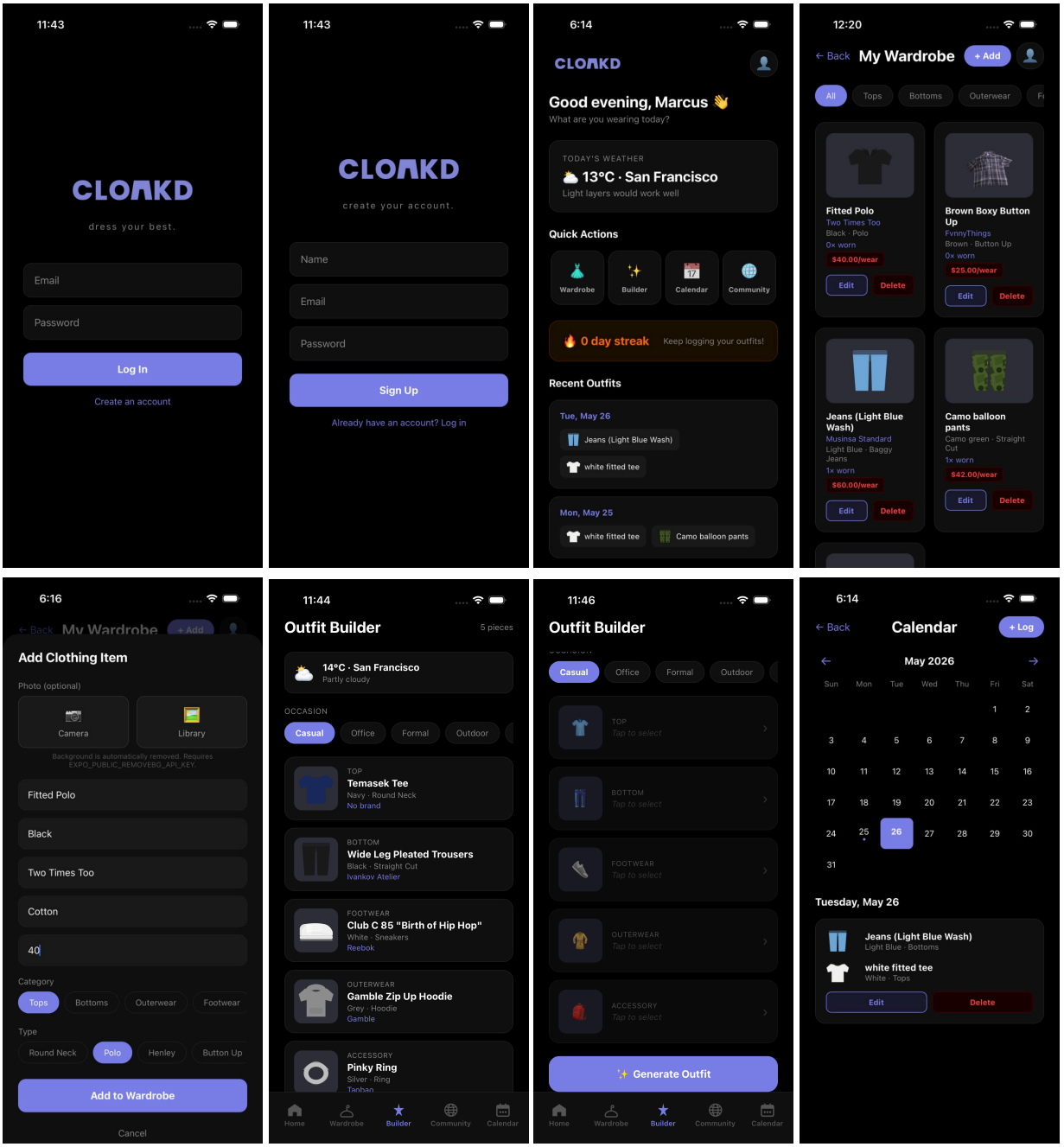
User Stories

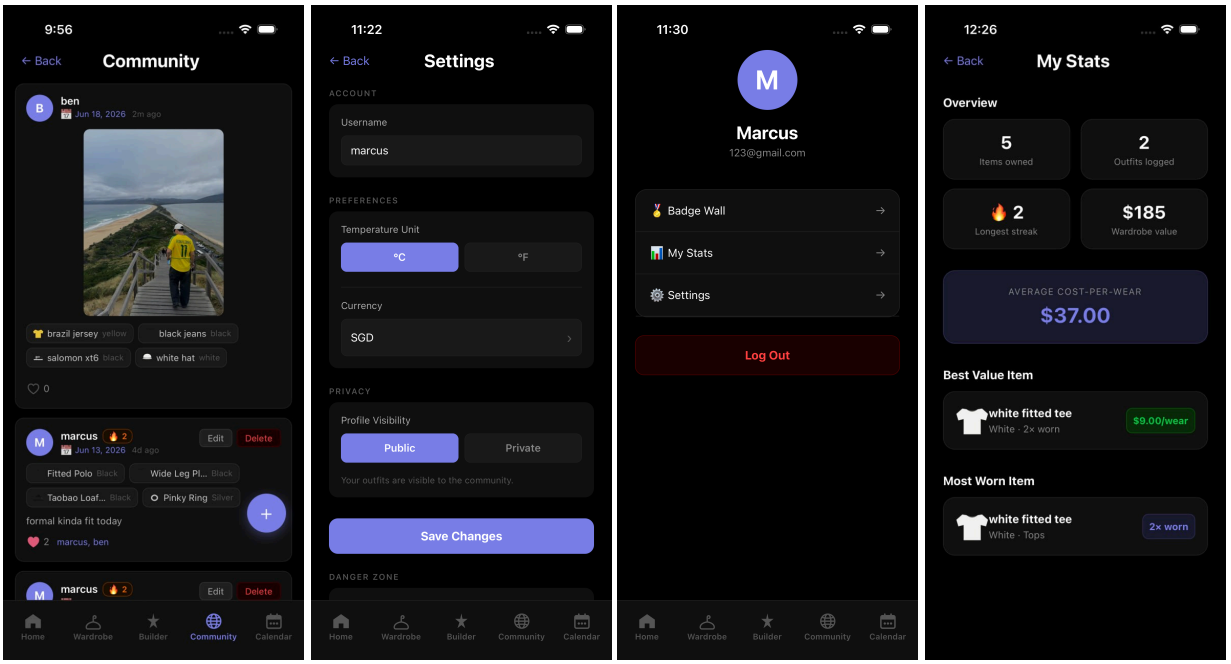
1. As a person who wants to dress better and improve my style, I want to be able to visualise and plan my outfits.
2. As a person who struggles with deciding what to wear, I want to be able to receive outfit suggestions for the day.
3. I want to be able to dress well and look stylish no matter the weather, even if it is a blazing hot day or a rainy day with thunderstorms. I also want to be able to dress appropriately for the occasion, such as date nights, formal occasions or gym days.
4. I want to be able to take inspiration from others' outfits and share my own.
5. As a frugal person, I want to be able to track how much value and how many wears I am getting out of my clothes.
6. I want to keep track of the outfits I have worn each day so that I do not repeat outfits too frequently.
7. As someone with a large wardrobe, I want to be able to search and filter my clothes by colour, category or material so that I can quickly find what I'm looking for without digging through my closet.
8. As someone who hasn't worn certain pieces in a long time, I want to be reminded of clothes I've forgotten about so that I can rediscover and make better use of them.
9. As someone who is trying to be more intentional about shopping, I want to see which gaps actually exist in my wardrobe before buying anything new.
10. As a conscious consumer, I want to understand the true cost of my purchases over time so that I can justify spending more on quality pieces that I'll wear repeatedly.
11. As someone who enjoys fashion, I want to browse trending outfits in the community and save looks that inspire me so that I have a personalised mood board to reference when planning my own outfits.
12. As someone who enjoys tracking personal progress, I want to look back at my outfit history over weeks and months so that I can see how my personal style has evolved.

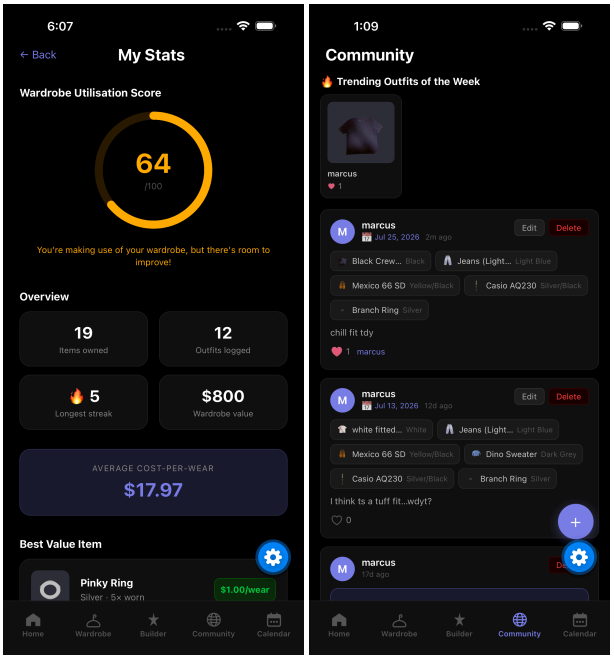
Scope of Project

CLOAKD is a mobile app that lets users photograph and catalogue every clothing item they own, then suggests daily outfits built entirely from their existing wardrobe — filtered by real-time weather and occasion. Users log outfits to a calendar, track cost-per-wear statistics, build daily streaks, earn achievement badges, and optionally share looks to a community feed.

Proof of Concept

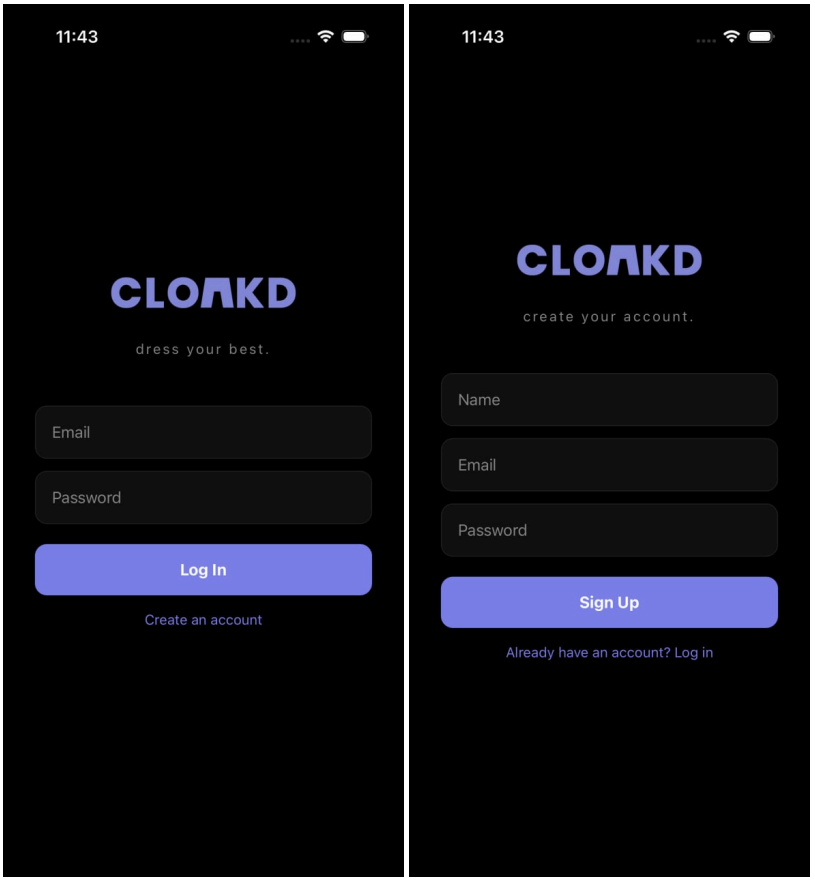



9:56
Community
ben
Jun 18, 2026 2m ago
brazil jersey yellow
black jeans black
salomon xt6 black
white hat white
marcus
Jun 13, 2026 4d ago
Edit
Delete
Fitted Polo Black
Wide Leg Pl... Black
Taobao Loaf... Black
Pinky Ring Silver
formal kinda fit today
2 marcus, ben
Home
Wardrobe
Builder
Community
Calendar
11:22
Settings
ACCOUNT
Username
marcus
PREFERENCES
Temperature Unit
°C
°F
Currency
SGD
PRIVACY
Profile Visibility
Public
Private
Your outfits are visible to the community.
Save Changes
DANGER ZONE
11:30
M
Marcus
123@gmail.com
Badge Wall
My Stats
Settings
Log Out
12:26
My Stats
Overview
5
Items owned
2
Outfits logged
2
Longest streak
$185
Wardrobe value
AVERAGE COST-PER-WEAR
$37.00
Best Value Item
white fitted tee
White · 2× worn
$9.00/wear
Most Worn Item
white fitted tee
White · Tops
2× worn


6:07
My Stats
Wardrobe Utilisation Score
64
/100
You're making use of your wardrobe, but there's room to improve!
Overview
19
Items owned
12
Outfits logged
5
Longest streak
$800
Wardrobe value
AVERAGE COST-PER-WEAR
$17.97
Best Value Item
Pinky Ring
Silver · 5× worn
$1.00/wear
1:09
Community
Trending Outfits of the Week
marcus
1
marcus
Jul 25, 2026 2m ago
Edit
Delete
Black Crew... Black
Jeans (Light... Light Blue
Mexico 66 SD Yellow/Black
Casio AQ230 Silver/Black
Branch Ring Silver
chill fit tdy
1 marcus
marcus
Jul 13, 2026 12d ago
white fitted... White
Jeans (Light... Light Blue
Mexico 66 SD Yellow/Black
Dino Sweater Dark Grey
Casio AQ230 Silver/Black
Branch Ring Silver
I think ts a tuff fit...wdyt?
0
marcus
17d ago
Home
Wardrobe
Builder
Community
Calendar

Features

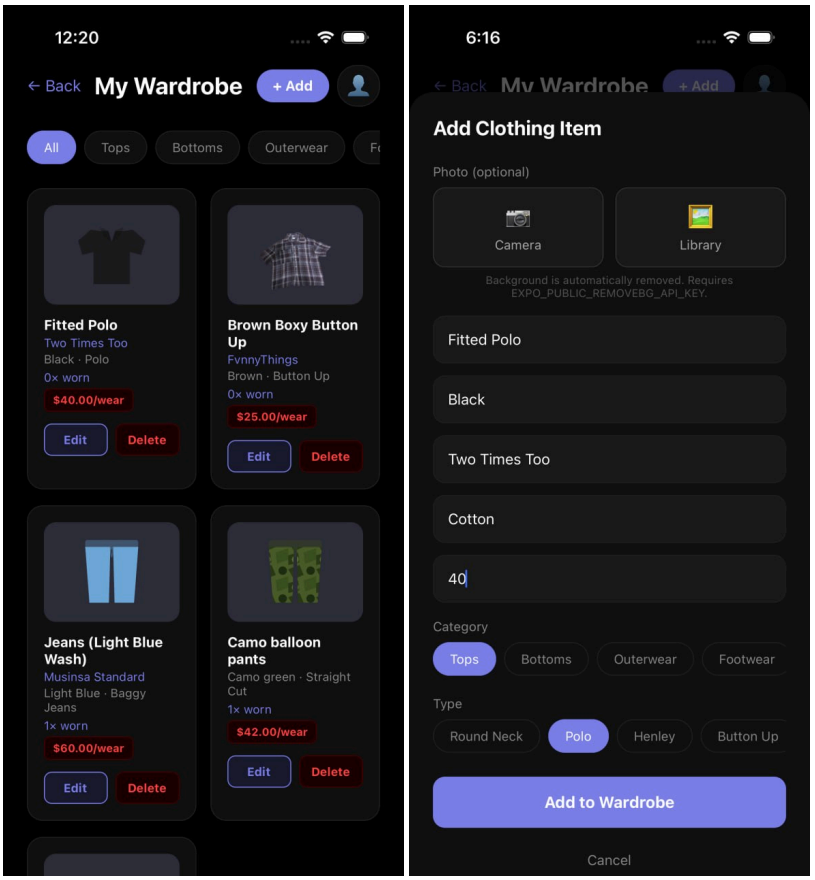
User Account Authentication



Users are authenticated with their own unique account via the Firebase SDK Authentication (email and password). FirebaseAuthExceptions are handled within a try-catch block and an alert is displayed when the user keys in an invalid input. Only upon successful login or account registration is the user routed to the home page.

Users can either log in with an existing account or sign up with their name, email and password. If account credentials are correct, they are redirected to the home page. Account deletion with password re-authentication is also fully supported.

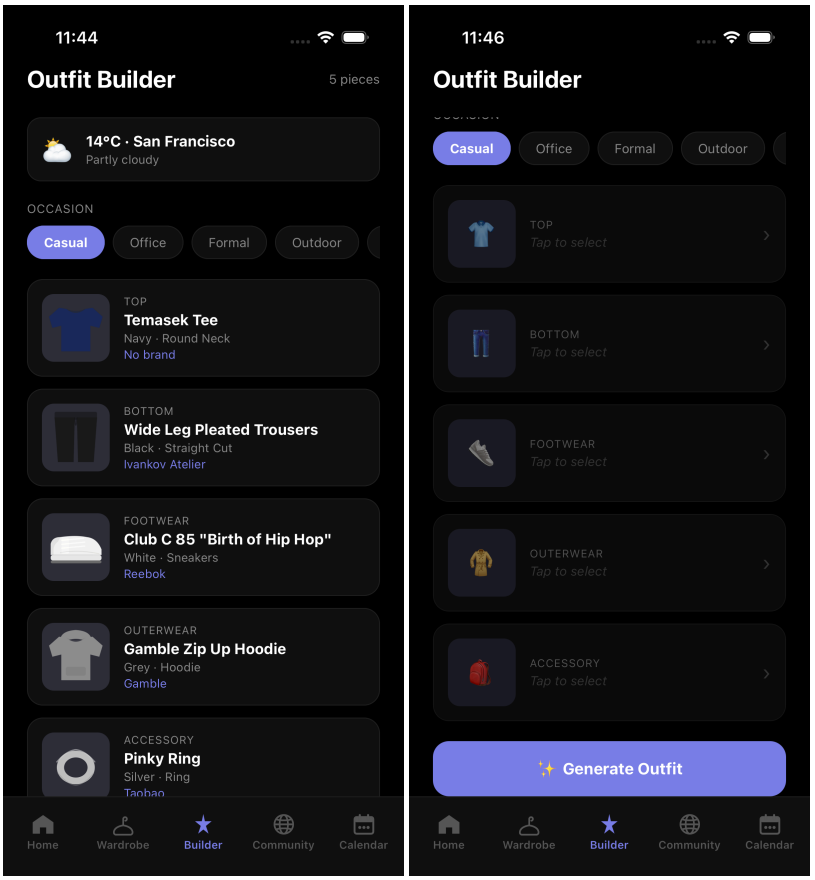
Virtual Wardrobe



Users add their clothes by uploading a photo of each item and tagging it with basic details such as colour, category, subcategory, material, brand and purchase price. The wardrobe screen displays all items at a glance, filterable by category, colour or subcategory in one tap. Items that have not been worn for 30 or more days and new items that have been added but are yet to be worn are surfaced automatically to encourage rediscovery.

Users can save clothing items with full metadata to Supabase. The wardrobe grid view displays all saved items with their uploaded photos. Image capturing with background removal is implemented. Filtering by category, colour and subcategory, as well as 30-day rediscovery surfacing, are both fully implemented.

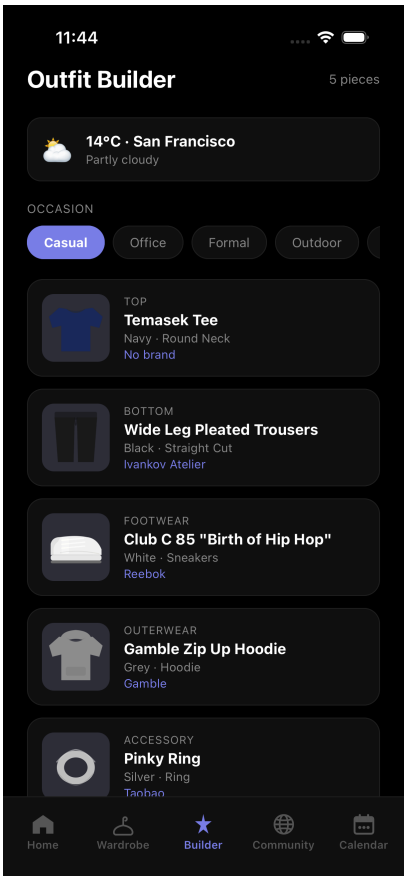
Outfit Builder



The outfit builder suggests outfit combinations built entirely from clothes the user already owns. The user is presented with a top, bottom, outerwear, footwear and accessories slot. Each slot is independently swappable, meaning the user can keep what they like and shuffle individual pieces. Weather and occasion filters are applied before suggestions load. Finalised outfits can be saved and logged.

Basic outfit builder feature has been created. The builder retrieves real-time weather data and prompts for occasion (casual, office, formal, outdoor, date night). Users can select each individual item manually or click the generate button to auto-build a colour-coordinated outfit from their wardrobe. Generating an outfit produces a layered visual preview so users can see how items combine before logging. The 30-day rediscovery logic and pushing of unworn pieces are fully integrated into the builder.

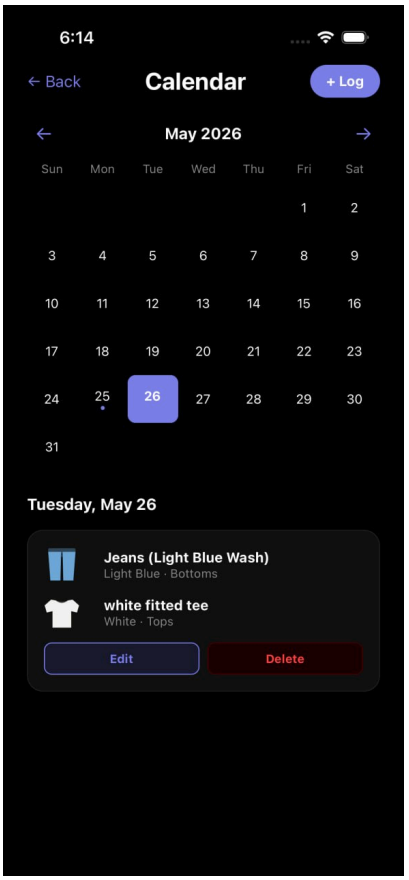
Occasion and Weather Filter



The filter asks the user where they are going and what they will be doing. It uses real-time weather data from the Open-Meteo API for the user's current location and filters the wardrobe to suggest outfits suitable for conditions and occasion. Supported occasions: casual, office, formal, outdoor, date night.

Open-Meteo API integration is complete and returns live weather data for the user's current location. The logic to filter wardrobe items by temperature range and occasion tag is connected to the outfit builder.

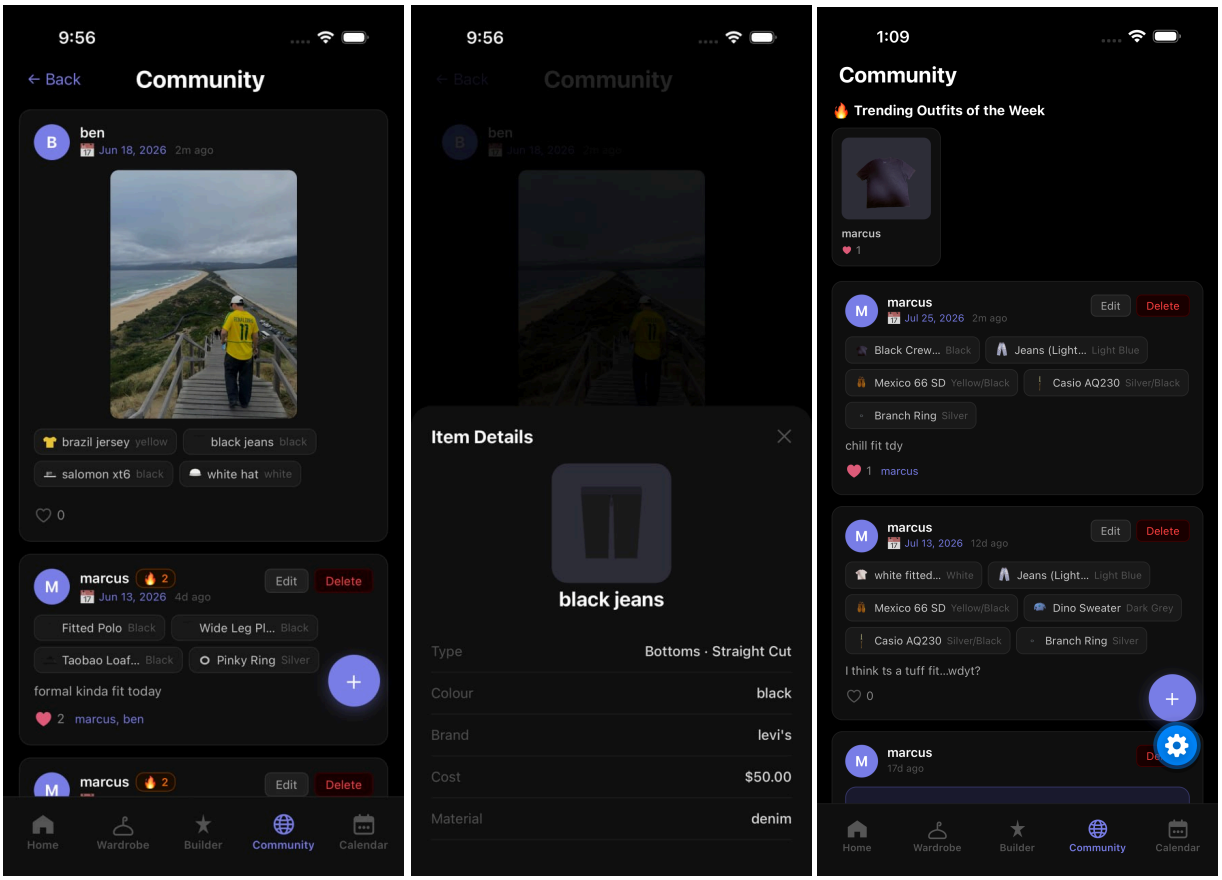
Outfit Calendar



The outfit calendar acts as a personal style diary. Every time a user logs an outfit, it is recorded against the specified date. Users can look back to see exactly what they wore and when. Logging an outfit automatically updates the wear count and cost-per-wear for every item in the outfit. Users can also edit and delete outfits that have been logged previously.

The calendar can be navigated by month. Logging an outfit to a date is functional and stored in the database. Automatic updating of wear counts, wear dates and cost-per-wear on log is complete. Visual indicators for logged days and the current date are displayed on the calendar grid.

Share Outfits and Community Feed



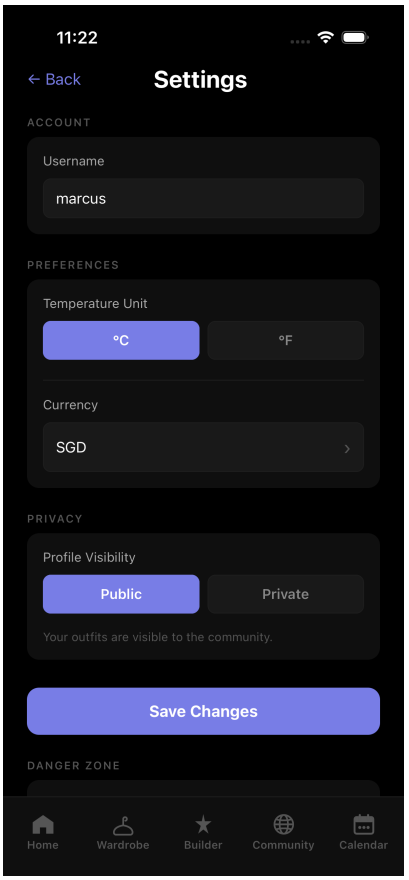
Users can opt in to post outfits to the CLOAKD community feed. Privacy is the default — a user's wardrobe stays invisible unless they choose to share. Shared posts display garment categories, colours and item photos. Other users can like posts and view who else has liked them.

The community feed is fully functional. Users can select a logged outfit, attach a photo and caption, and share it to the community feed. Each post displays the outfit date, caption, shared photo, and the individual clothing items as tappable chips with their wardrobe photos. Tapping a clothing chip opens a detail panel showing the item's name, category, subcategory, colour, brand, purchase price and material.

The liking system is complete. Users can like and unlike any post with a single tap. Like counts update immediately with optimistic UI and revert automatically on error. Users can tap the like count to see a list of everyone who has liked the post.

When a user updates their username in Settings, it propagates automatically across all their existing posts and likes in the community feed.

Settings Page

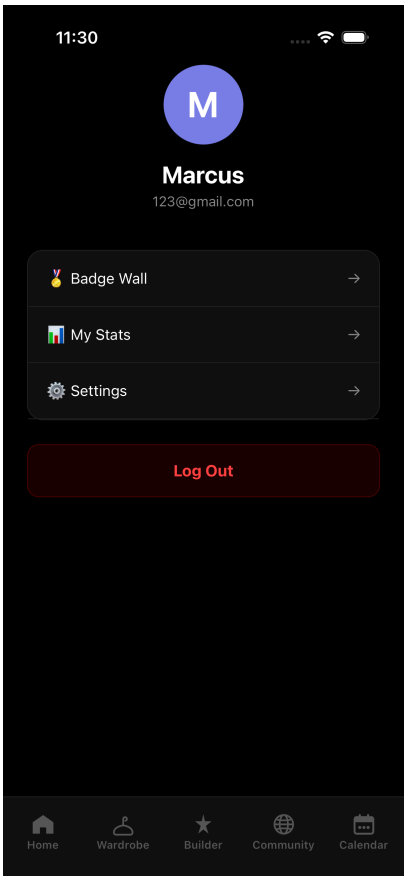


A dedicated settings screen lets users manage their account details and app preferences in one place. Users can set a display username, choose their preferred temperature unit, select their currency, and control the visibility of their profile. Account deletion is available as a permanent action behind a password confirmation step.

The settings page is complete. Users can:

- **Username** — set a display name (minimum 3 characters, unique) that appears on all community posts and likes. Changing the username automatically updates it across all existing posts and likes in real time.
- **Temperature Unit** — toggle between °C and °F. The selection is persisted and reflected immediately in the home page weather card.
- **Currency** — select from SGD, USD, EUR, GBP, JPY, AUD, CAD, HKD, MYR, or IDR for cost-per-wear display.
- **Profile Visibility** — toggle between Public (outfits visible to the community) and Private (outfits visible to the user only).
- **Delete Account** — permanently deletes the account and all associated wardrobe data after password re-authentication. This action is irreversible.

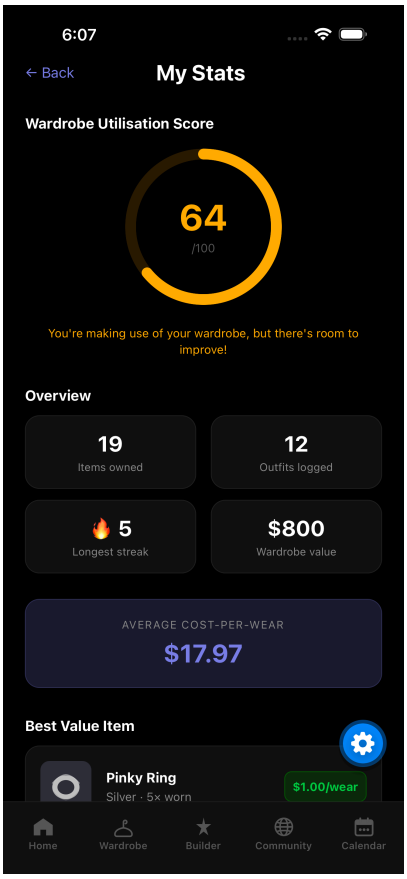
Profile Page



Each user has a profile page that displays their username, profile photo, email address, badge wall, stats and settings. The profile is accessible from the home page and serves as a personal hub for reviewing personal style history and achievements.

The profile page is accessible from the profile button on the home page header. The user can view their username, email address and profile photo, and navigate to the stats page, settings page and badge wall. The user can also log out from this page.

Statistics Page

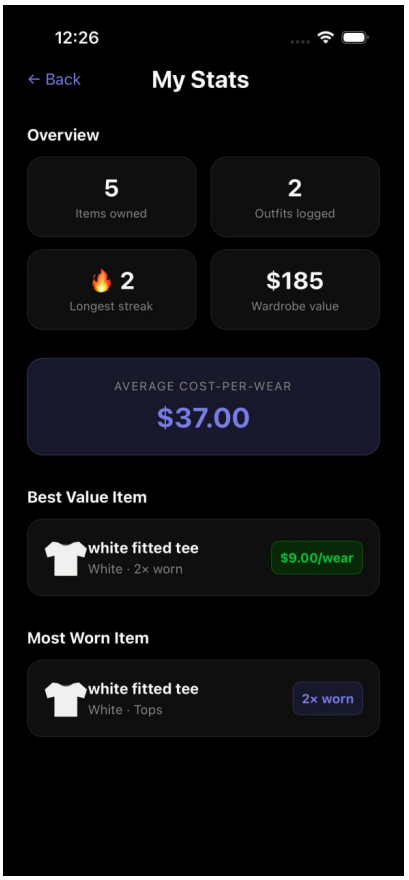


Users can view their account statistics on the statistics page. It contains figures such as their Wardrobe Utilisation Score, total number of items owned and outfits logged, longest streak, total wardrobe value, average cost-per-wear for all the clothing items in their wardrobe, as well as their best value and most worn items.

Users can also see their total wardrobe value and the total number of clothing items they own if they are interested to see how much they have spent on their wardrobe as a whole.

The average cost-per-wear figure shows the average cost-per-wear of all the clothing items in a user's wardrobe and also helps to encourage them to wear their clothing items more and log outfits to get better value out of the clothing items they already own.

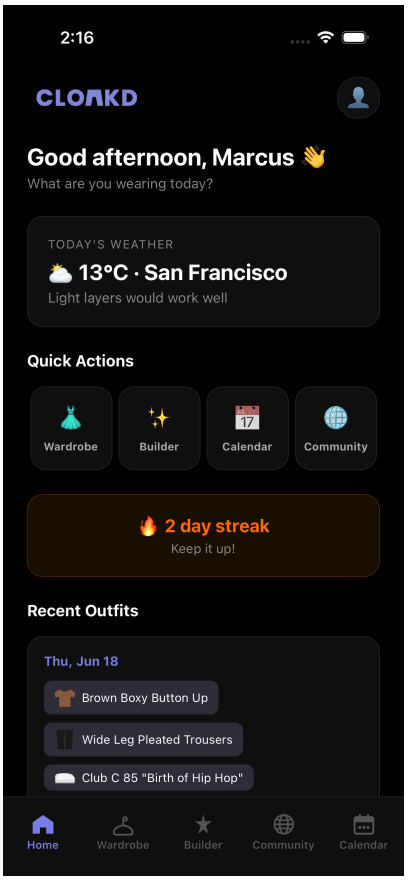
Cost-Per-Wear Tracker



When adding clothing items, users input the purchase price. Cost-per-wear is calculated as price divided by the number of wears and updated automatically each time the item is logged as part of an outfit. The wardrobe view surfaces this figure alongside wear count for each item.

The cost-per-wear tracker is fully built. Users can see the number of times they have worn each piece and its cost-per-wear in the wardrobe page. Logging an outfit in the calendar automatically updates the wear count and cost-per-wear. The stats page shows the best value item (lowest cost-per-wear), average cost-per-wear across the wardrobe, and the most worn item.

Streak System

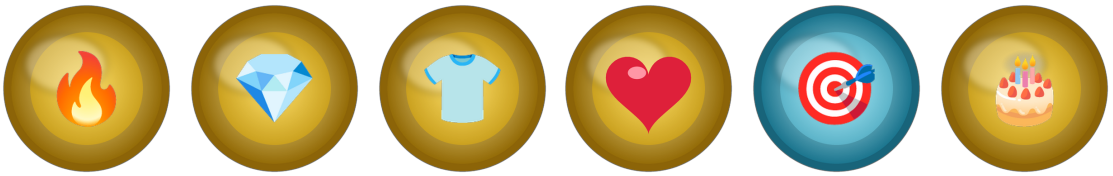


Every day a user logs an outfit, their streak grows. Missing a day resets the counter. Badges unlock at 7-day, 30-day and 100-day milestones to keep users motivated to open the app daily.

The streak system is fully implemented. A streak is activated when a user logs outfits on 2 or more consecutive days. Missing a day resets the streak to 0. The streak counter is live on the home page with motivational subtext. Users with no active streak are prompted to log 2 days in a row to start one.

The streak is also visible in the community feed: when a user with an active streak (2+ days) shares an outfit, a fire icon and streak count appear beside their username on the post card, visible to all community members.

Badge Wall

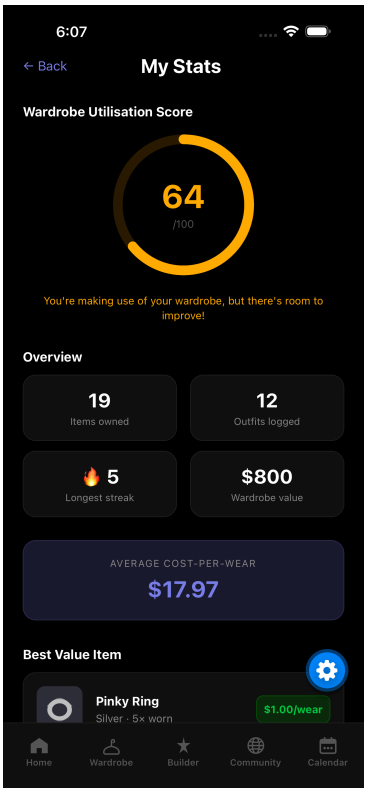


The badges page is complete with 36 badges. Users automatically unlock badges once they meet each badge's requirements and can share newly unlocked badges directly to the community feed.CLOAKD's badge system rewards users for consistent engagement across every core feature, such as logging streaks, tracking cost-per-wear, growing their wardrobe and being active in the community. Badges are organised into 11 categories, with progression tiers representing increasing milestones within each category. Progress bars on each badge card show real-time completion, and badges are automatically unlocked and persisted once their condition is met.

Category	Badge	Tier	Unlock Requirement
Getting Started	Welcome to Cloakd	Bronze	Create an account
Getting Started	Early Adopter	Bronze	Log an outfit within your first week
Getting Started	First Post	Bronze	Share your first outfit to the community
Streaks	Warming Up	Bronze	Reach a 7-day streak
Streaks	In Rotation	Silver	Reach a 30-day streak
Streaks	Unkillable Habit	Gold	Reach a 100-day streak
Cost-Per-Wear	Getting Your Money's Worth	Bronze	An item's cost-per-wear drops under $5
Cost-Per-Wear	Basically Free	Gold	An item's cost-per-wear drops under $1
Wardrobe Size	Just Getting Started	Bronze	Log 10 wardrobe items
Wardrobe Size	Building the Archive	Silver	Log 25 wardrobe items
Wardrobe Size	Stacked Inventory	Gold	Log 50 wardrobe items
Outfits Logged	One Day or Day One	Bronze	Log 1 outfit
Outfits Logged	Creature of Habit (the good kind)	Silver	Log 10 outfits
Outfits Logged	Century Club	Gold	Log 100 outfits
Community	Conversation Starter	Bronze	Get 3 likes on a single outfit post
Community	Getting There	Silver	Get 10 likes on a single outfit post
Community	Fashionista	Gold	Get 40 likes on a single outfit post
Community	Pinterest Wall	Diamond	Get 100 likes on a single outfit post
Community	Fan Favourite	Silver	Have the same user like 5 of your posts

Category	Badge	Tier	Unlock Requirement
Style Diversity	Rainbow Dresser	Silver	Wear 5 different colours in one week
Style Diversity	Brand Hopper	Silver	Wear 5 different brands in one week
Outfit Recycler	Old Faithful	Bronze	Wear the same item 10 times
Outfit Recycler	Wardrobe Warrior	Gold	Wear the same item 50 times
Outfit Recycler	Vintage Lover	Silver	Log an item that's been in your wardrobe for 1+ year
Wardrobe Health	Investment Piece	Silver	Add an item worth $200 or more
Wardrobe Utilisation	Awakened	Bronze	Reach a utilisation score of 50+
Wardrobe Utilisation	Wardrobe Aware	Silver	Reach a utilisation score of 75+
Wardrobe Utilisation	Full Rotation	Gold	Reach a utilisation score of 90+
Wardrobe Utilisation	Maximalist	Diamond	Reach a perfect utilisation score of 100
Wardrobe Utilisation	Comeback	Silver	Improve utilisation score by 20+ points in one month
Wardrobe Utilisation	Rising Tide	Silver	Improve utilisation score for 2 consecutive months
Wardrobe Utilisation	Well Dressed, Well Used	Gold	Maintain a utilisation score of 75+ for 30+ days
Wardrobe Utilisation	No Piece Left Behind	Diamond	Maintain a utilisation score of 90+ for 30+ days
Anniversary	First CLOAKDaversary	Bronze	1 year since account creation
Anniversary	CLOAKD Legend	Silver	5 years since account creation
Anniversary	CLOAKD GOAT	Gold	10 years since account creation

Wardrobe Utilisation Score



The wardrobe utilisation score gives users a single number that answers the question: how much of what you own are you actually wearing? It sits at the top of the Stats page with its own dedicated header and acts as the headline metric for wardrobe health.

The score is calculated as the percentage of wardrobe items that are **not forgotten**, rounded up to the nearest whole number, capped at 100. An item is considered forgotten if it has never been worn, or if its most recent wear was more than 30 days ago — the same definition used by the wardrobe rediscovery section.

> Score = ⌈ (items worn in the last 30 days ÷ total items) × 100 ⌉, capped at 100

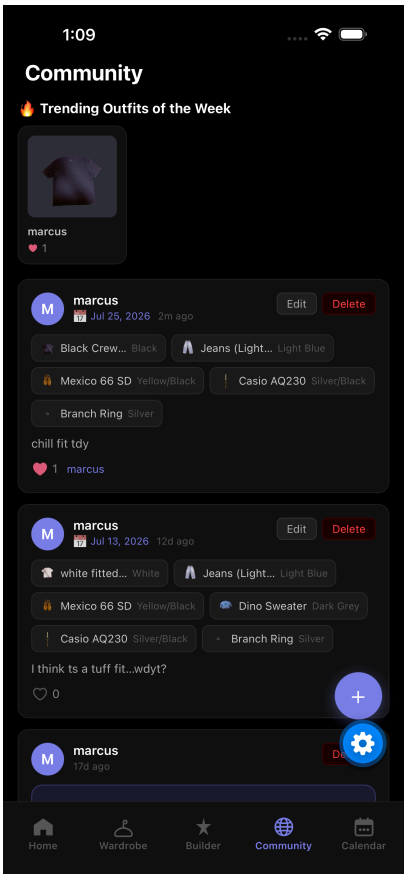
The score updates automatically whenever an outfit is logged or an item is added or deleted, since both events change either the numerator (items with a recent wear) or the denominator (total items).

The score and its progress ring are colour-coded by tier:

Score Range	Colour	Tooltip
0 – 50	Red	"You have many unworn clothes, try giving them a chance!"
51 – 75	Yellow	"You're making use of your wardrobe, but there's room to improve!"
76 – 90	Green	"Great work! Most of your wardrobe is being put to good use."
91 – 100	Purple	"Outstanding! Your wardrobe utilisation is exceptional!"

The UI renders the score inside a circular progress ring drawn with `react-native-svg`. The ring arc is filled proportionally to the score (0 = empty, 100 = full circle). Both the arc and the central score label change colour together according to the active tier.

Trending Outfits



On the community feed page, users share their outfits and like other users' shared outfits. A horizontal scroll section has been created at the top of the community feed page to showcase the trending outfits of the week. The formula to calculate the most trending outfits is likeCount / (ageInHours + 2)^1.5 and it helps to ensure that a new post with fewer likes can outrank an older post with more likes, keeping the section feeling more current and up to date with trends rather than just all-time popular outfits.

Only outfit posts within the last 7 days with at least one like are eligible to be featured in this section and only the top 10 outfits by score will be showcased. The score is automatically recomputed and the outfits are updated whenever posts or like counts change.

If a user has set their account to private, their outfits will not be featured in this section even if they are less than 7 days old and have at least one like.

Testing

Testing Approach

CLOAKD is tested through a combination of manual end-to-end testing, structured feature walkthroughs, and unit tests targeting edge cases in core business logic. Given the app's reliance on real-time data from Firebase and Supabase, integration testing is prioritised to ensure the full data flow works correctly in production conditions. Unit tests focus on pure functions (cost-per-wear calculation, streak logic, colour matching) that can be isolated from the network layer.

Unit Tests Edge Cases

The following edge cases are covered by unit tests on the core logic functions.

Cost-Per-Wear Calculation

CPW is computed as `price / wear_count` when `wear_count > 0`, and falls back to `price` when `wear_count === 0`. The CPW badge is only rendered when `price > 0`.

Test Case	Input	Expected Output
Item never worn, has price	price = $120, wears = 0	Displays $120.00/wear (full price used as CPW — no division by zero)
Item worn once	price = $120, wears = 1	Displays $120.00/wear
Item worn many times	price = $50, wears = 1000	Displays $0.05/wear (rounds to 2 decimal places)
Free item (price = $0)	price = $0, wears = 5	CPW badge not shown (hidden by `price > 0` guard)
Price with many decimal places	price = $99.999, wears = 3	Displays $33.33/wear (`.toFixed(2)` applied)
Item with no price set	price = $0, wears = 0	CPW badge not shown

Streak Calculation

A streak requires the most recent logged date to be either today or yesterday. Dates are deduplicated and sorted before counting. A streak is only reported when 2 or more consecutive days are found.

Test Case	Input	Expected Output
No outfits logged	dates = []	Streak = 0
Single outfit logged today	dates = [today]	Streak = 0 (need 2+ consecutive days)
Two consecutive days	dates = [today, yesterday]	Streak = 2
All consecutive from today	dates = [today, d-1, d-2, d-3]	Streak = 4
Gap in streak	dates = [today, yesterday, d-3, d-4] (d-2 missing)	Streak = 2 (chain breaks at the gap)
Most recent date is yesterday, not today	dates = [yesterday, d-2, d-3]	Streak = 3 (yesterday counts as anchor)
Most recent date is 2+ days ago	dates = [d-2, d-3, d-4]	Streak = 0 (too stale to count)
Dates out of order	dates = [d-3, today, d-1, d-2]	Streak = 4 (sorted correctly before calculation)
Duplicate date entries	dates = [today, today, yesterday]	Streak = 2 (deduplicated before calculation)

Outfit Colour Matching

Colour compatibility is determined by family groups (red/burgundy/maroon, blue/cobalt/sky blue, etc.) and a neutral set (white, black, grey, beige, navy, etc.). Neutrals are compatible with everything. After 200 failed attempts to build a harmonious outfit, the generator falls back to the first available item in each pool regardless of colour.

Test Case	Expected Output
All items are neutrals	Any combination is accepted as harmonious
Item colour not in any family and not neutral	Treated as incompatible with any non-neutral; fallback outfit used after 200 attempts
Wardrobe has only one item per slot	Fallback always fires; that item is used regardless of colour match
All items filtered out by weather and occasion (no tops or bottoms remain)	Full unfiltered wardrobe used as fallback — no empty state
Occasion filter leaves no tops or bottoms	Full unfiltered wardrobe used as fallback

Username Validation

Minimum length is 3 characters (checked via `.trim().length < 3`). Uniqueness is enforced via a Supabase unique constraint (error code `23505`). There is no maximum length enforced in the app. Internal spaces are permitted.

Test Case	Input	Expected Output
Exactly 3 characters (boundary)	"abc"	Accepted
2 characters (below minimum)	"ab"	Rejected — "Username must be at least 3 characters" alert
Empty string	""	Rejected — same alert
Username already taken	existing username	Rejected — "That username is already in use" alert
Username with internal spaces	"my name"	Accepted — spaces are not stripped or rejected
Very long username	100-character string	Accepted — no maximum length is enforced

Weather Filter

The weather filter uses four temperature thresholds (°C): above 28, 22–28, below 18, below 10. Each threshold blocks specific subcategories rather than requiring specific ones. If applying the filter leaves the wardrobe with no tops or no bottoms, the full unfiltered wardrobe is used instead.

Test Case	Temperature	Expected Behaviour
Hot (> 28°C)	35°C	Heavy outerwear (jacket, coat, puffer, blazer, windbreaker), sweatpants, hoodies and crewnecks excluded
Warm (22–28°C)	25°C	Only heavy outerwear (coat, puffer) excluded
Mild (18–22°C)	20°C	No items excluded
Cool (< 18°C)	15°C	Sandals, shorts, and light dresses excluded
Cold (< 10°C)	5°C	All dresses and all exposed-top subcategories (tank, crop, halter, etc.) also excluded
Filter leaves no tops or bottoms	Any	Full unfiltered wardrobe used as fallback — no empty result
Location permission denied	N/A	Weather card shows "Location unavailable"; builder defaults to 22°C (no alert or prompt shown)

Manual Test Cases

User Authentication

Test	Expected Result	Status
Sign up with a new email and password	Account created, redirected to home page	Pass
Sign up with an email already registered	Error alert displayed, no account created	Pass
Sign up with a password under 6 characters	Error alert displayed	Pass
Log in with valid credentials	Redirected to home page	Pass
Log in with incorrect password	Error alert displayed, no redirect	Pass
Log in with unregistered email	Error alert displayed	Pass
Delete account with correct password	Account and data removed, redirected to login	Pass
Delete account with wrong password	Error alert, account retained	Pass
Session persists after app close	User remains logged in on reopen	Pass

Virtual Wardrobe

Test	Expected Result	Status
Add a clothing item with all fields	Item saved and visible in wardrobe grid	Pass
Add a clothing item with only required fields	Item saved with blank optional fields	Pass
Add a clothing item with photo	Photo displayed in wardrobe grid and builder	Pass
Add a clothing item without a photo	Clothing icon placeholder shown	Pass
Edit an existing clothing item	Updated details reflected immediately	Pass
Delete a clothing item	Item removed from wardrobe and any logged outfits	Pass
Filter wardrobe by category	Only matching items shown	Pass
Filter wardrobe by colour	Only matching items shown	Pass
Filter wardrobe by subcategory	Only matching items shown	Pass
Wardrobe with zero items	Empty state message shown, no crash	Pass
30-day rediscovery section	Items unworn for 30+ days surfaced at top	Pass
New unworn item surfaced	Newly added but never-worn items appear in rediscovery	Pass

Outfit Builder

Test	Expected Result	Status
Generate an outfit	Items selected from wardrobe matching weather and occasion	Pass
Manually swap a slot	Selected item replaces the slot	Pass
Log a generated outfit	Outfit saved to calendar on the selected date	Pass
Generate outfit with empty wardrobe	Empty state prompt shown, no crash	Pass
Generate outfit with wardrobe missing one category	Builder fills available slots, leaves missing slot empty	Pass
Outfit preview renders correctly	Layered clothing preview shown in correct order	Pass
Preview updates when slot is swapped	Preview refreshes immediately on item change	Pass

Outfit Calendar

Test	Expected Result	Status
Log an outfit to a date	Date highlighted on calendar, wear counts updated	Pass
Log an outfit to a past date	Accepted and stored correctly	Pass
Log a second outfit to the same date	Replaces or appends correctly, no duplicate	Pass
View a logged outfit	Items displayed correctly for that date	Pass
Navigate between months	Calendar updates without data loss	Pass
Delete a logged outfit	Date indicator removed, wear counts decremented	Pass

Streak System

Test	Expected Result	Status
Log outfits on 2 consecutive days	Streak of 2 shown on home page	Pass
Skip a day after an active streak	Streak resets to 0	Pass
Log multiple outfits on the same day	Streak does not double-count	Pass
Streak badge on community post	Fire icon and count appear beside username	Pass
Streak refreshes on home tab focus	Updated count shown after returning from calendar	Pass
No streak when only one day logged	"Log 2 days in a row to start a streak" prompt shown	Pass

Community Feed

Test	Expected Result	Status
Share an outfit post	Post appears at top of community feed	Pass
Share a post with photo	Photo displayed in post card	Pass
Share a post without photo	Post displays with item chips only, no crash	Pass
Like a post	Like count increments immediately	Pass
Unlike a post	Like count decrements immediately	Pass
Like reverts on network error	Count restored to pre-tap state	Pass
View likers list	All usernames who liked the post are shown	Pass
Tap clothing chip in feed	Item detail modal opens with correct data	Pass
Tap clothing chip in fullscreen post	Item detail popup opens over fullscreen view	Pass
Tap item card in fullscreen post	Detail popup with item photo and metadata appears	Pass
Edit own post	Caption and photo updated in feed	Pass
Delete own post	Post removed from feed	Pass
Feed scrollable with many posts	Scroll functions correctly, no layout freeze	Pass
Post with no items	Post renders cleanly with no chip section	Pass

Settings

Test	Expected Result	Status
Change username	New username reflected on all existing posts and likes	Pass
Set username under 3 characters	Error shown, username not saved	Pass
Set duplicate username	Error shown, username not saved	Pass
Switch temperature unit to °F	Home page weather updates to Fahrenheit	Pass
Switch temperature unit back to °C	Home page weather updates back	Pass
Change currency	Currency symbol updates in cost-per-wear displays	Pass
Currency setting persists on app reopen	Correct currency shown after closing and reopening app	Pass
Set profile to private	Posts hidden from community feed	Pass

Cost-Per-Wear

Test	Expected Result	Status
Item with 0 wears	Displays "—" or equivalent, no division by zero error	Pass
Log outfit containing item	Wear count increments, CPW recalculates	Pass
Stats page — best value item	Item with lowest CPW shown correctly	Pass
Stats page — most worn item	Item with highest wear count shown correctly	Pass

Devices Tested

Device	OS	Method
iPhone 17 (Simulator)	iOS 18	Xcode iOS Simulator

> Testing was conducted on a simulated iPhone 17 via Xcode rather than a physical device. This is because the version of Expo Go available on physical devices had not been updated to support the SDK version used in this project.

Known Limitations

- Testing is currently limited to the iOS Simulator. Physical device and Android compatibility testing has not been completed.
- Weather data accuracy is dependent on the Open-Meteo API and device location permissions being granted. Location is simulated in the Xcode Simulator and may not reflect real-world GPS accuracy.
- Unit tests are described and verified manually against isolated function behaviour. A formal test runner (e.g. Jest) has not been integrated into the project build pipeline.

Timeline and Development Plan

Milestone	Date	Task & Description
Liftoff	10 - 18 May 2026	Create project poster.
Liftoff	10 - 18 May 2026	Film, edit and craft liftoff project poster.
Liftoff	10 - 18 May 2026	Finalise project features and review proposal.

> **Evaluation Liftoff: 18 May 2026**
>
> - Create project poster and video.
> - Solidify plans for CLOAKD app.
> - Self-learn all the tools in our tech stack to build our CLOAKD app.

Milestone	Date	Task & Description
1	19 - 24 May 2026	Set up Expo project, Firebase backend and Supabase backend. Integrate Expo with Firebase for user authentication and Supabase for clothing item information.
1	19 - 24 May 2026	Create log-in and sign-up pages with user email and password for authentication and verification.
1	19 - 24 May 2026	Create home page as a dashboard that is connected to all other pages and features.
1	24 - 31 May 2026	Create virtual wardrobe feature with image capture and input of clothing item details such as colour, material, subcategory and purchase price. Link to Supabase for storage of details.
1	24 - 31 May 2026	Create outfit calendar feature with ability to log outfits and highlight dates where outfits have been logged.
1	24 - 31 May 2026	Create Cost-Per-Wear extension to update wear count every time the clothing item is logged in an outfit. Use purchase price / number of wears to calculate.

> **Evaluation Milestone 1: 1 June 2026**
>
> - Able to add clothes to the virtual wardrobe + store and access data.
> - Outfit Calendar functional.
> - Cost-per-wear calculation functional. Integration with Virtual Wardrobe complete.
> - Account sign-up and log-in with authentication complete.
> - Basic UI complete.

Milestone	Date	Task & Description
2	2 - 15 June 2026	Create outfit builder feature and link it to the virtual wardrobe to directly access all clothing items.
2	2 - 15 June 2026	Add function that pushes clothes that have not been worn in a while for outfit builder feature.
2	2 - 15 June 2026	Include occasion and weather filter to suggest outfit recommendations for outfit builder feature.
2	16 - 28 June 2026	Create settings page.
2	16 - 28 June 2026	Build community feed and outfit sharing feature.

> **Evaluation Milestone 2: 29 June 2026**
>
> - Occasion and Weather Filter functional.
> - Outfit builder functional.
> - Virtual Wardrobe backend complete. Integration with Cost-per-wear Tracker done.
> - Community feed frontend complete.

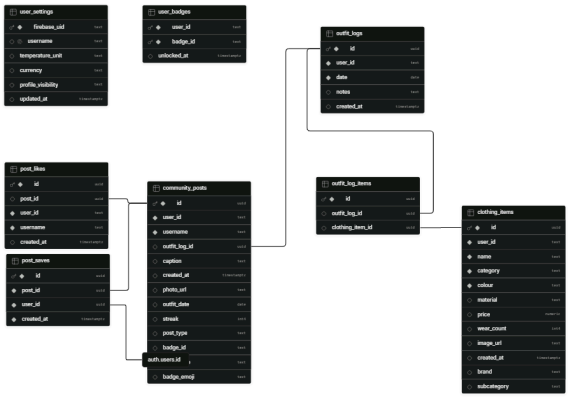
Milestone	Date	Task & Description
3	1 - 26 July 2026	Complete streak system.
3	1 - 26 July 2026	Create achievement badges for streaks, outfits logged, cost-per-wear milestones and wardrobe size.
3	1 - 26 July 2026	Ensure community feed is fully functional end-to-end.
3	1 - 26 July 2026	Complete outfit sharing feature.
3	1 - 26 July 2026	Outfit preview layered visual.

> **Evaluation Milestone 3: 27 July 2026**
>
> - Streak system functional with badges and achievement tokens.
> - Share Outfits feature fully functional.
> - Community feed fully functional.
> - Outfit preview complete.

Milestone	Date	Task & Description
Splashdown	1 - 25 August 2026	Fix all bugs and improve UI.
Splashdown	1 - 25 August 2026	Make quality of life updates and improvements.

> **Evaluation Splashdown: 26 August 2026**
>
> - Fix all bugs and improve UI.
> - Quality of life updates and improvements.

Database Schema



Tech Stack

Layer	Technology
App Framework	React Native, Expo
Navigation	Expo Router (file-based tab navigation)
PostgreSQL Database	Supabase
File Storage	Supabase Storage (community post images)
Authentication	Firebase Authentication (email + password)
Weather API	Open-Meteo
Location	expo-location
Image Capture	expo-image-picker
Custom Icons	react-native-svg
Version Control	Git, GitHub

Project Log

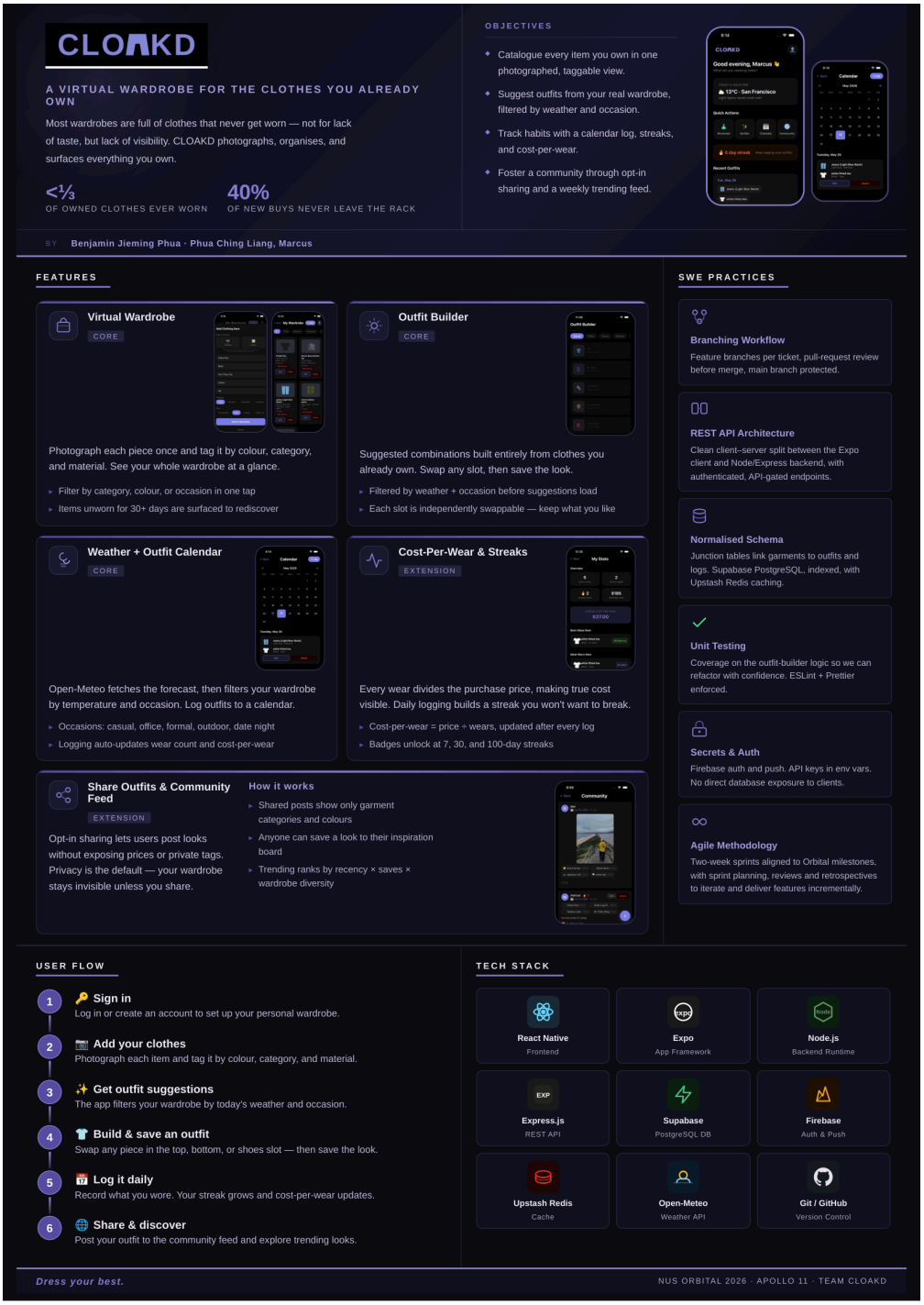
Project Log

Project Video

Project Video

Project Poster

Project Poster



Dress your best. · *Team CLOAKD*

Benjamin Jieming Phua · Phua Ching Liang, Marcus